

Anik Ghosh

Mobile : +91 8240676812

| avickg@gmail.com | <https://www.linkedin.com/in/anik-ghosh-443a7b1b0/>
<https://github.com/AnikGhosh332?tab=repositories>

PROFILE

I have 2.5+ years of experience as a machine learning engineer with specialisation in building and deploying production-grade **ML** and **Generative AI** systems, with a strong focus on transformer-based models, LLM fine-tuning, NLP pipelines, and MLOps. Experienced in end-to-end system development, applied machine learning, data pipelines, and user-facing analytical tools, with a strong interest in skills intelligence, workforce analytics, and applied research-led innovation.

EDUCATION

University of Liverpool - Master of Science Data Science and AI

Sept 2022 - Sept 2023

- *Core topics : Python Programming, Maths and Statistics, Data mining and visualisation, Supervised learning, deep learning, Neural Networks, Natural Language Processing, Computer Vision and Dissertation*
- Degree Classification: **Distinction**

Vellore Institute of Technology - Bachelor of Technology Electronics and Communications Engineering

July 2017 - June 2021

- *Core topics : Computer Networks, Digital Electronics, Signal Processing, MATLAB, Python, Data structures and algorithms, Mathematics (Statistics and Linear Algebra), Operating Systems, Analog Electronics and C/C++ Programming*
- Graduated with an overall **CGPA of 8.59/10**

WORK EXPERIENCE

Machine Learning and Data Engineer | Valuechain Technology Ltd

April 2024 to Feb 2025

- Automated data extraction using Selenium and built an ETL pipeline using Apache Airflow, ensuring efficient data ingestion and transformation. The data was stored using Amazon Redshift.
- Designed and implemented a recommendation engine using transformer models that improved tender matching accuracy by 25% and deployed it using Docker for scalable production deployment.
- Built and deployed a production-ready web application using Django, with AWS S3 for file storage, Tailwind CSS for the front-end, and hosted on Heroku for rapid deployment.
- Worked with cross-functional teams and analysed user behaviour using Mixpanel to identify drop-off points and engagement trends, leading to data-driven product enhancements and improve user retention rates by 15% and created data visualisations using PowerBI.
- Designed an LLM based email intelligence system to classify and extract actionable intents from emails. This was done using Gmail API and fine-tuning a transformer model, enabling automated downstream workflows.
- Used MLflow and Prometheus to deploy and monitor ML models in production. Applied MLOps practices like CI/CD (GitHub Actions), DVC-based version control, and automated retraining to improve accuracy and ensure model reliability over time.

Associate Software Engineer | Victoria's Secret & Co

Nov 2021 to Sep 2022

- Redesigned and enhanced an internal stock management tool using Node.js, Express, and Oracle SQL, improving database integration and overall user experience, resulting in smoother inventory tracking and faster data updates.
- Developed data pipelines using Python which would gather shipment data and generate detailed reports. The data was stored in MS SQL.
- Developed and automated several test cases using Selenium and Behaviour Driven Development (BDD) using Cucumber which reduced the overall functional test completion time by 30%.
- Gained hands-on experience with test automation tools including Playwright and Detox, and developed automated API tests for backend services.
- Configured CI/CD pipelines to run Selenium tests automatically using Jenkins/GitHub Actions, reducing manual effort and improving test reliability.

Programmer Analyst | Cognizant Technology Solutions

Feb 2021 to May 2021

- Successfully completed training on web technologies such as HTML, CSS, Javascript, and Azure Cloud Platform and received outstanding feedback on my final presentation and project.
- Developed and optimised SQL queries and collaborated with developers to improve data processing efficiency.
- I developed a Client Issue Tracking and Resolution System using C# and ASP.NET MVC.

Front End Web Developer | Ericsson India

May 2020 to Jul 2020

- Developed a single page web application for maintaining a list of company employees and their profile.
- Used HTML, CSS and Javascript to build the web application and used Angular as the front end framework
- Fundamental rules of UI/UX were implemented to make the website readable, navigable and accessible.

SKILLS

- **C++, Java, Python (Pytorch, Pandas, Tensorflow, Scikit Learn, Langchain, Matplotlib), C#**
- **Automation Testing, Manual Testing, Selenium, Unit Testing, UI Testing, A/B Testing**
- **HTML, CSS, React, Node.js , TypeScript, Javascript, RESTful API, FastAPI, Django**
- **MYSQL, Oracle SQL, SQLite, MongoDB, SQL Server, Database Design**
- **Data Visualisation (Matplotlib and Power BI), Data Mining, Data Modelling , Machine Learning (supervised and unsupervised learning) , Statistical Analysis, Deep Learning, Natural Language Processing and Large Language Models**
- **GIT, Apache Airflow, Docker, Apache Spark, Kubernetes, Amazon Web Service, Azure, Linux, CI/CD**

PROJECTS

Human Activity Recognition using Limited Training Data

- Developed strong expertise in **multivariate wearable sensor time-series analysis**, including multi-resolution feature extraction across temporal scales, subject-level variability handling, and human-participant data considerations.
- Designed and implemented zero shot learning, few shot learning and transfer learning models on human activity recognition using PAMAP2 dataset enabling systematic comparison in scenarios with limited training data.
- Implemented prototype-based few-shot learning and fine-tuned transfer learning models, demonstrating substantial performance gains over zero-shot approaches on unseen activities.
- Conducted rigorous, leakage-free evaluation using subject-disjoint splits, confusion matrices, and collapse diagnostics, highlighting real-world deployment challenges in wearable sensor ML.

EMNIST dataset image classification

- Developed and trained MLP and CNN model using Python to classify the given handwritten characters to their respective labels.
- The model's performance was analysed and improved through hyper-parameter tweaking.

Job Recommendation Engine

- Designed a NLP pipeline using Retrieval-Augmented Generation (RAG) to recommend relevant jobs from a large corpus of job descriptions based on semantic similarity with a candidate resume.
- Generated vector embeddings for resumes and job descriptions using **Sentence-BERT**, and applied similarity scoring and ranking to identify best-matching roles.
- Integrated an LLM via API to generate structured natural language summaries explaining match strengths and gaps between a resume and each recommended job.
- Designed the system as an exploratory project to evaluate the feasibility of using **GenAI** for resume screening and job matching.

Flight Booking Website

- Built a full-stack flight booking web app using Node.js, Express.js, MongoDB, and Bootstrap to simulate an airline reservation system.
- Implemented features for searching flights, and performing CRUD operations on bookings (create, view, update, cancel).
- Designed a RESTful API backend with modular routing, middleware, and Mongoose-based data models for maintainability.
- Followed best practices in code organisation, error handling, and basic input validation to ensure a clean, scalable architecture.

YouTube Trending Video Analysis & Ad Performance Prediction

- Utilised a Kaggle dataset to analyse trending YouTube videos and identify the types of videos best suited for successful ad campaigns.

- Built an end-to-end data pipeline on AWS using Amazon S3, AWS Glue, and Athena to ingest, transform, and analyse a Kaggle dataset of trending YouTube videos.
- Developed machine learning models (Logistic Regression, Random Forest, XGBoost) to predict ad suitability and engagement potential, using features derived from video metadata and early performance signals.
- Created an interactive Streamlit dashboard to visualise trending patterns, model predictions, and feature importance, and deployed the application on AWS EC2 using Docker.

ACHIEVEMENTS

- Lead my team in the hack-a-thon for smart building designs and communicated my idea within my team members and presented it to the panel and managed to secure 11th position among 327 participating teams.
- Recipient of scholar badge award for outstanding performance in 11th grade and 12th Higher Secondary Board Exam.